

the rocks of the Saanich granodiorite formation exists at Sharp point¹ about 50 miles to the northwest. Unless igneous activity has taken place more recently than late Jurassic the existence of a spring such as this is difficult to explain.

It is possible that a spring similar to that at Sharp Point may have at one time existed near this deposit of mercury and may have deposited the cinnabar. It is known that cinnabar is at present being precipitated by the waters of many hot springs in California and Nevada, such, for example, as those at Sulphur Bank, California, Steamboat Springs, Nevada, and Rabbit Hole Sulphur deposit, Humboldt county, Nevada. It is found that mercury is precipitated by springs whose waters carry sodium chloride, sodium carbonate, and sodium borate, and the Sharp Point spring is of the sodium chloride type.

It is difficult to judge the size of the deposit from the exceedingly poor exposures that exist. For a distance of about 200 yards along the creek small patches of very lean ore of no commercial value are promiscuously scattered in the country rock. Near the south end of the showings is a large dump containing several thousand tons of ore taken from a shaft that could not be examined on account of the water which filled it, but the ore on the dump consisted of impregnations of cinnabar and native mercury in andesite, limestone, and breccia. Some of the specimens of impregnated andesite were quite rich, but the limestone and breccia formed only lean ore. A sample of the dump taken across its height assayed 0.38 per cent mercury.

It is the writer's opinion that with thorough prospecting a commercial deposit of mercury might be discovered in this vicinity.

SUNLOCH COPPER DISTRICT, B.C.

By Victor Dolmage.

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INTRODUCTION.

The Sunloch copper deposit, situated on Jordan river, Vancouver island, was discovered a few years ago by George E. Winkler of Victoria, who, after doing the necessary development work, had the claims crown-granted and in 1917 bonded them to R. H. Stewart and associates of Vancouver. A company was organized called the Sunloch Mines, Limited, which is now developing important ore-bodies on the property.

The mine, though still in an early stage of development, promises to become eventually an important producer of copper. Its development is of peculiar interest to the people of British Columbia since it is one of the very few mining ventures which are being successfully developed by a local company of comparatively small investors.

¹ Clapp, C. H., Geol. Surv., Can., Sum. Rept., 1913, p. 80.

The situation of the property with reference to transportation, power, water supply, and easy accessibility, is almost ideal. The workings are located on the steep banks of Jordan river, 2 miles above its entrance into the strait of Juan de Fuca and only 40 miles west of Victoria. It is connected with the beach by a narrow-gauge tram-line running down to the mouth of the canyon, and a standard gauge logging railway which terminates at a wharf near the mouth of the river. The hydro-electric generating plant of the Vancouver Island Power Company, capable of supplying ample power for all purposes, is located within 2 miles of the tunnels. A good motor road, over which is maintained a tri-weekly stage service, runs from Victoria to the settlement near the mouth of Jordan river. The river furnishes a convenient and ample water supply for milling purposes.

GEOLOGY.

The geology of the district surrounding the Sunloch mineral deposit is similar to that of the western half of the Sooke sheet, which lies not more than 2 miles east of the workings on the Sunloch claims. The geology of the Sooke sheet has been worked out in detail and fully described by C. H. Clapp,¹ and C. H. Clapp and H. C. Cooke².

According to these authors, three formations comprise the geology of this part of Vancouver island, the Metchosin volcanics of lower Tertiary, the Sooke gabbros of middle Tertiary, and the Sooke formation of upper Tertiary age. The Sunloch and adjacent mineral deposits are at or near the contacts of a mass of Sooke gabbro which intrudes the Metchosin volcanics.

The Sooke formation which consists of partly consolidated, finely bedded sediments unconformably overlying the other two formations, does not outcrop on the area described in this report. It does, however, outcrop a few hundred yards to the south of Sinn Fein creek which crosses the Black Hornet claim and enters Jordan river just below the Vulcan claims.

The distribution of the other two formations in the vicinity of the claims is shown on Figure 1. The gabbro mass extends eastward, probably beyond the boundary of the Sooke sheet, and it is highly probable that it may be connected, at no great distance below the surface, with one of the gabbro masses which outcrop a short distance east of the Sooke sheet boundary. This is an important consideration with reference to the probable size of the Sunloch ore-bodies, because this small mass of gabbro, which is concluded to be the source of the ore, might be expected to produce a much larger quantity of ore if it were an off-shoot of some nearby larger mass of gabbro and not a small independent and isolated mass.

Metchosin Formation.

Distribution and Extent. Of the three formations the Metchosin is much the largest, occupying a belt from 5 to 7 miles wide, which extends along the southwest coast of Vancouver island from Albert head to Lost river, a distance of 35 miles. Except for two small remnants of the Sooke sediments, and five or six small intrusions of Sooke gabbro, the Metchosin volcanics are the only rocks outcropping in this area.

Lithology. This volcanic series includes all the usual types, such as amygdaloids, agglomerates, flow breccias, and tuffs, but all are distinctly basic in composition. The normal basalt consists of labradorite ranging from Ab₃₀ An₇₀ to Ab₄₀ An₆₀, and augite with accessory magnetite and apatite. It varies in texture from microcrystalline to medium coarse-grained. It commonly has the diabasic structure well

¹ Clapp, C. H., Geol. Surv., Can., Mem. 13, 1912.

² Clapp, C. H., and Cooke, H. C., Geol. Surv., Can., Mem. 96, 1917. Cooke, H. C., Geol. Surv., Can., Mus. Bull. 30, 1919.

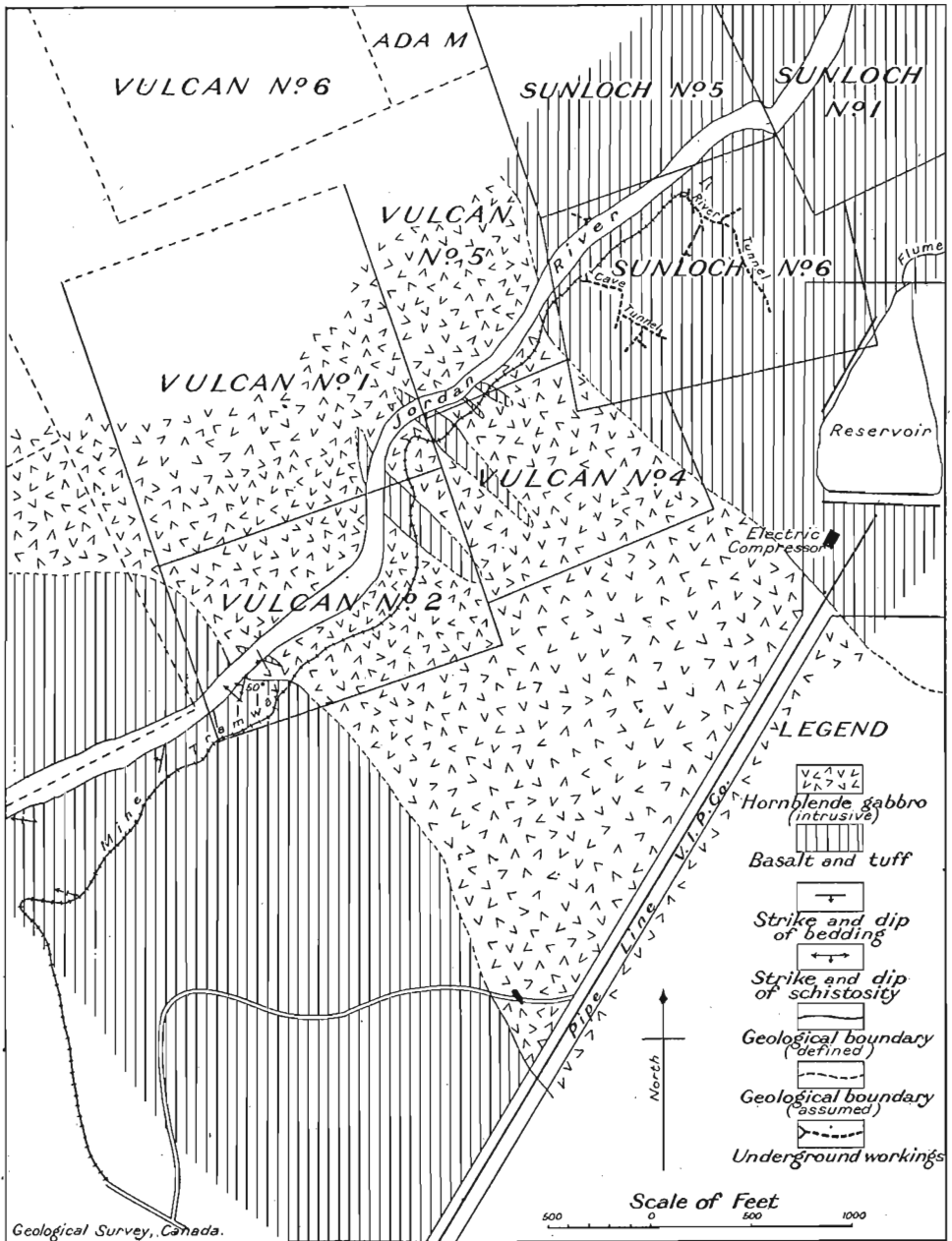


Figure 1. Sunloch copper deposits, Jordan river, Vancouver island, British Columbia.

developed, and in some places it is distinctly porphyritic. The coarse-grained varieties strongly resemble the gabbro intrusions. The flows are found in places to be cut by diabase dykes which are clearly related to the volcanic types, and are identical with many of the basalts.

Metamorphism. The alteration of these basic lavas and tuffs has been slight. Only the less stable minerals, such as the ferromagnesian, have undergone chemical alteration. Even this alteration has not been intense, and consists in their replacement by small amounts of chlorite and serpentine.

Although generally only slightly altered, the Metchosin volcanics have been intensely contact-metamorphosed in the neighbourhood of the intrusions of Sooke gabbro, and this is particularly true of the basalts and tuffs exposed on the Sunloch and Vulcan mineral claims. Clapp¹ says of this contact metamorphism:

"The contact zones are seldom more than 200 to 300 feet in width and are frequently narrower, although in a few places, especially along sheared zones, the contact-metamorphosed and mineralized areas are found at distances of a mile or more from any outcrop of the intrusives. Where less metamorphosed the original texture of the basalt is partially preserved since the feldspars remain unaltered and form diversely to subradially arranged laths set in a fine-grained lustrous groundmass which is clearly seen. . . . to consist largely of hornblende. On microscopic examination the original ophitic structure is conspicuous, but the augite is in most rocks replaced by a light sage to brownish green, weakly pleochroic fibrous to almost compact hornblende. The brownish and more compact hornblende is very similar to that developed in the Sooke gabbro itself, and in a few places forms large poikilitic crystals. In the more metamorphosed rocks the original ophitic texture is less conspicuous and irregular areas of actinolite and chlorite, as well as of uraltite have been developed, and veinlets and replacements of epidote and zoisite are common. The feldspars remain unchanged. The more metamorphosed rocks have been impregnated with finely crystalline pyrite, pyrrhotite, and chalcopyrite and also replaced and cut by veinlets of these minerals. These metallic minerals are seldom abundant except in sheared zones in the contact-metamorphosed rocks. Some of the shear zones are, however, highly metamorphosed and in them are numerous prospects. The sheared and mineralized rocks consist largely of aggregates of secondary minerals, chiefly hornblende, actinolite, epidote, chlorite, quartz, and secondary feldspar, cut by veinlets of quartz and epidote. The metallic minerals have developed late in the metamorphism of the rocks, and occur disseminated through the earlier secondary minerals, and in veinlets and irregular replacements, usually associated with quartz, cutting the earlier quartz epidote veins."

Metchosin Basalts in the Vicinity of the Sunloch Deposits. In the vicinity of the Sunloch and adjoining claims the Metchosin volcanics are both sheared and intruded by a considerable mass of gabbro and are in consequence metamorphosed to such a degree that the original character of the rocks is almost obliterated. In their original condition, no doubt, they were similar to many of the unaltered types found in the Sooke area and described briefly in a preceding paragraph, but all the volcanic rocks examined, and included in the accompanying map, are now metamorphosed by contact and dynamic action. They are composed of greenish and brownish hornblende, labradorite, occasionally augite and small amounts of magnetite, pyrite, chalcopyrite, and veinlets of quartz, epidote, zoisite, chlorite, calcite, and sericite.

Of these minerals the hornblende is the most abundant, forming from 50 per cent to 90 per cent of the rock. Two varieties were observed, a greenish brown hornblende, giving pleochroic shades of brownish green to yellowish green, and a deeper green variety, giving shades from light green to deep bluish green, the latter often forming veinlets which cut the brownish hornblende, as well as the labradorite and other minerals. The brownish hornblende is closely associated with small veins of

¹ Clapp, C. H., Geol. Surv., Can., Mem. 96, p. 270.

augite and bears the same relation to the feldspars as the augite does in the less altered basalts. It is, therefore, believed to have been formed from the alteration of primary augite. The feldspar forms from 5 to 20 per cent of the specimens examined and occurs both as laths and well developed tabular crystals, which are more than ordinarily inclined to idiomorphism. It ranges in composition from rather basic labradorite to bytownite, the former predominating. It is usually characterized by very fine twinning lamellæ, formed by both the albite and pericline laws. It is comparatively little altered, but in places was observed to be replaced by sericite, chlorite, quartz, and actinolite. In the basalts of this vicinity augite is an extremely rare constituent, and in all the basalts that were collected in the neighbourhood of the mine only a few scattered grains of augite were noted, due probably to its alteration to hornblende. These were either small interstitial grains or larger allotriomorphic crystals enclosing blades of labradorite. All the secondary minerals, excepting the hornblende, occur in very small amounts. Quartz is on the whole the most abundant of the minerals, but is confined altogether to the vicinity of quartz veins and areas of basalt that have been extensively sheared. In these areas it is very abundantly disseminated in certain light coloured, gneissic-looking bands. Epidote is not so plentiful as one would expect in such highly altered volcanic rocks and is seen only in very small veins associated with quartz and sulphides. The sulphides—chalcopyrite, pyrrhotite, and pyrite—are sporadic in their occurrence and irregularly distributed. They are, however, largely confined to areas in the vicinity of the gabbro contacts that have undergone considerable shearing, and it is in such areas that the commercial ore-bodies of these deposits are developed.

Structure. The Metchosin formation consists of a series of superimposed lava flows and beds of tuff ranging in thickness from 2 or 3 feet to 300 feet. As is always the case in such volcanic series the conformity existing between the beds is by no means perfect and many large irregularities occur in the planes of contact. Owing to such irregularities and to the difficulty of distinguishing the various flows, because of lithological similarity and metamorphism, the structure of the formation is not easily determined. In places, however, where cherty tuffs are exposed, or where the lava flows are thin, folding was observed and is found to have taken place along axes which strike north 60 degrees to 70 degrees west. The prevailing dips are to the northeast, and Clapp¹ cites evidence which leads him to the conclusion that the part of this formation occupying the Sooke sheet forms a part of the southern limb of a geosyncline.

Besides this general folding on a large scale, local intense folding is observable, which is thought to be confined to the more incompetent beds. These minor folds have strikes approximating the direction of the main folds, but having more variable dips.

Shearing has been extensively developed in the Metchosin basalts along the great fault which bounds them on the north and to a lesser degree in the vicinity of the gabbro intrusives. As will be pointed out later, the shearing has an important bearing on the development of the mineral deposits of the district.

Sooke Gabbro.

General Character and Distribution. The gabbro mass exposed in the vicinity of the Sunloch and Vulcan mineral claims is considerably altered and its original character to a great extent obliterated, but in its present state of metamorphism it very strikingly resembles the metamorphosed parts of the Sooke gabbro and has the same relation to the Metchosin basalts.

This formation has been described in great detail by H. C. Cooke², of the Geological Survey.

The formation exposed in the Sooke map-area consists of five small bodies, the largest of which, occupying the East Sooke peninsula, forms an elliptical mass about

¹ Clapp, C. H., Geol. Surv., Can., Mem. 96.

² Cooke, H. C., "Gabbros of East Sooke and Rocky point," Geol. Surv., Can., Mus. Bull. 30, 1919.

5 miles in length. The only known occurrence outside the Sooke and Duncan map-area is the one on the Sunloch claims, situated about 3 miles west of the boundary of that area.

These small stocks, besides being sheared and highly altered by mineralizing solutions, have undergone considerable magmatic differentiation, with the result that they now present a great variety of rock types. Cooke describes the four following types: olivine gabbro, augite gabbro, anorthosite, and granite, and it is probable, from its resemblance, that the Sunloch gabbro is a representative of the olivine gabbro in a state of intense alteration.

The olivine gabbro, as described by Cooke, is a medium, coarse-grained, grey to black rock, consisting approximately of 45 per cent to 50 per cent of bytownite feldspar, about the same amount of augite, 5 per cent of olivine, and 1 per cent of ilmenite, but so irregular in composition that within short distances the above proportions may vary between wide limits. In its altered condition it carries from 50 per cent to 100 per cent of hornblende.

The Sunloch meta-gabbro, an altered phase of the Sooke gabbro, is a black, irregularly grained rock, consisting of 10 per cent to 45 per cent labradorite-bytownite feldspar, 50 per cent to 100 per cent hornblende, a very small amount of augite, and a few grains of ilmenite, magnetite, and titanite. No olivine was observed in the material collected. The almost total absence of augite is due to its alteration to hornblende, a change which took place before the rock was noticeably fractured and invaded by mineralizing solutions. The result of this early action on the augite is the development of a pale green, fibrous hornblende, strongly pleochroic from pale green to yellowish green. Several grains of augite in a partly altered state were observed. At a much later period the gabbro was strongly fractured and attacked by solutions which deposited great quantities of greenish brown hornblende as veinlets cutting the feldspars, augite, and the pale green hornblende. Another variety of hornblende, of yellowish brown colour, probably titaniferous, is thought to be a primary constituent.

Structure. The gabbro masses intrusive into the Metchosin basalts and exposed in the Sooke map-area, are elongated along two parallel axes which conform to the direction of the main folding, i.e., north 60 degrees to 70 degrees west. The projection of the southern axis in a westerly direction beyond the boundary of the Sooke map-area would pass close to the Sunloch claims. It is, therefore, probable that the gabbro mass exposed on the Sunloch claims is a part of the large body presumably underlying the greater part of the Sooke map-area and with which the other stocks outcropping in the area are connected in depth. The Sunloch mass, in so far as it has been mapped, has two nearly parallel boundaries and would, therefore, be classed as a dyke rather than a boss or stock. Its northern contact, as exposed in the canyon of Jordan river, has a nearly vertical dip to the north. The southern contact is also nearly vertical, but more irregular than the northern.

ORE DEPOSITS.

Distribution. Copper deposits occur on or near both contacts of the Sunloch gabbro dyke. About 300 feet beyond the northeast contact are several ore-bodies of the Sunloch group. On the southwestern contact where it crosses the river are several showings belonging to the Vulcan group of claims, and where this contact reaches Sinn Fein creek are the showings of the Black Hornet claim.

The Vulcan and Black Hornet deposits had not, up to the time of examination, been developed to any extent, and little can be said concerning them, except that the surface showings are strong and worth exploring.

Structure of the Ore Deposits. A recognition of the structure obtaining in the country rocks is an important need in development work. The significant feature, which is shown on Figure 2, is: the bedding planes within a few miles of the deposit strike from north 60 degrees west to north 90 degrees west and dip from 60 degrees to 90 degrees in a northerly direction.

The probable trend of the axes along which the gabbro has been intruded is parallel to the main axis of folding in the basalts. The Sunloch gabbro, however, in the vicinity of the deposits, strikes north 50 degrees west and, therefore, cuts the bedding planes at an angle of about 30 degrees.

A pronounced schistosity has been developed in the basalt close to and parallel to the contacts of the gabbro. The basalts are cut also by a series of small faults striking north 10 degrees east to north 40 degrees east which are, therefore, nearly at right angles to the strike of the bedding planes. These faults are all small with throws of not more than a few feet and were mostly, if not all, formed prior to the periods of ore deposition.

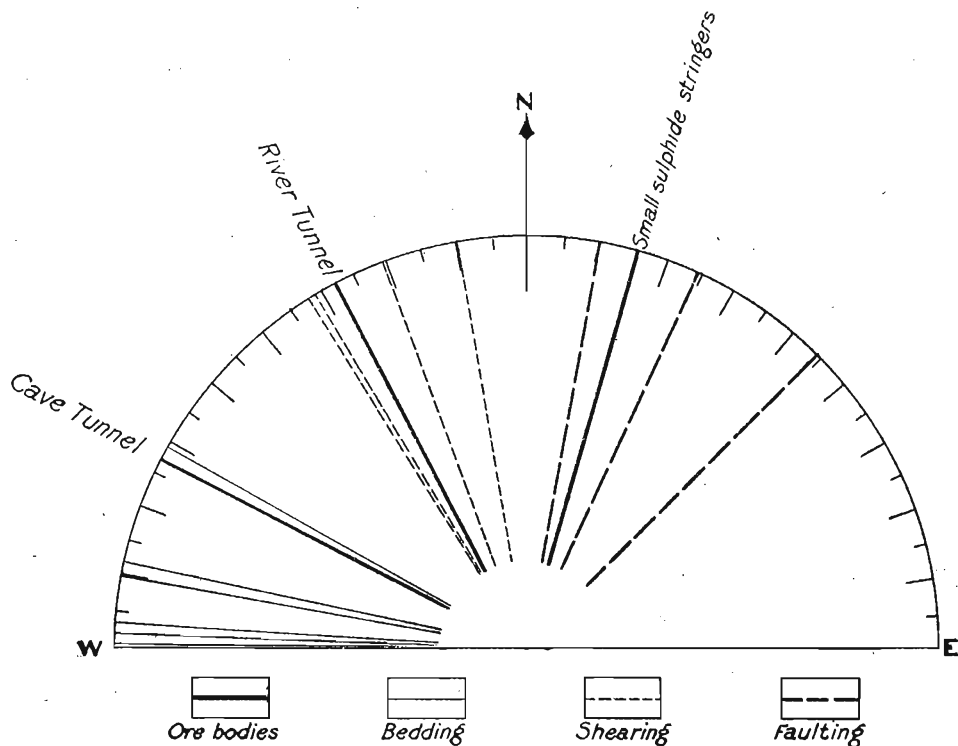


Figure 2. Diagram showing the relationship of the ore-bodies to the bedding, shearing, and faulting, Sunloch copper deposits, Jordan river, Vancouver island, B.C.

A persistent set of closely spaced joints was observed in the tunnels, one group of which strikes north 30 degrees west and dips 80 degrees to the northeast, and the other group strikes north 55 degrees east and dips 83 degrees to the northwest.

Since the planes of schistosity are the most pronounced lines of weakness in the rocks they have provided the easiest channels for ore solutions and the largest and most promising ore-bodies are consequently developed in them.

The planes of bedding are another determining factor in the location of smaller deposits than those formed in the planes of schistosity; but because of the irregularity of these planes the ore-bodies formed in them are more irregular in size and direction. Such a deposit is that exposed in the Cave tunnel. A third set of ore-shoots, which are scarcely more than stringers, follows directions somewhat parallel to the faulting, but the stringers are too small to be of any value.

A projection of the two main ore-bodies along the strike indicates an intersection at a distance of about 1,000 feet and if mineralization occurs at this point, an ore-

body larger than any yet discovered may be present, which would strike in the direction of the river zone and pitch to the northwest at an angle greater than 80 degrees.

Some of the more competent beds of volcanic rock are not sheared. These beds are devoid of mineralization and it is one of these competent beds which defines the northern boundary of the river zone. It has a steep dip to the north and conforms to the dips of the other beds in the vicinity.

The Ore. In hand specimens the ore consists of a sheared and hornblenditized basalt containing either a large proportion of finely disseminated chalcopyrite, or a smaller amount of chalcopyrite distributed in a network of small, filmy veinlets. Veinlets of either quartz and epidote carrying pyrite and chalcopyrite, or aplitic material, are occasionally seen. Alternating layers of rich and lean ore produce a banded appearance in the deposit. The ore is invariably finely granular, and large masses of pure sulphides are rarely seen. Small grains of pyrite are plentifully distributed throughout the ore.

The composition of the Sunloch ore is simple. The following is a complete list of the minerals, named in order of deposition: magnetite, hornblende, aplite, quartz, epidote, pyrite, molybdenite, chalcopyrite, limonite, and chalcocite.

The magnetite is rare and occurs both in the ore and country rock, of which it is probably a primary constituent.

The hornblende is very abundant and was introduced at two different periods, the first soon after the intrusion of the gabbro when the augite was completely replaced by hornblende, and the second at a much later time and after some, at least, of the chalcopyrite had been deposited. This later hornblende is a deeper green variety associated with considerable actinolite and appears in the form of minute veinlets cutting the earlier hornblende as well as the pyrite and chalcopyrite.

The small aplite stringers were formed during the early period of alteration and are cut by other veinlets carrying quartz, epidote, pyrite, and chalcopyrite.

The pyrite is in general much earlier than, and extensively replaced by, the chalcopyrite; but a small amount was introduced at a very late date in the periods of mineralization and occurs in the form of minute stringers of pyrite cutting all the other minerals excepting the limonite and chalcocite. Such late stringers of pyrite are, however, rare in ore of this class.

The deposition of the chalcopyrite was spread over a long and uninterrupted period, during which small quantities of molybdenite were also deposited.

The limonite is due to oxidation at a recent period and is found only in small amounts usually associated with pyrite rather than with chalcopyrite.

The chalcocite is probably of secondary origin and of a very recent period. It occurs in quantities much too small to have any commercial influence on the value of the ore.

Sunloch Ore-bodies.

On the Sunloch property there are three ore zones roughly parallel and approximately 300 feet apart. The most southerly, namely, the Cave ore zone, is about 400 feet north of the gabbro contact; the most northerly is the River zone, and between these lies the Archibald showing.

The River zone is the most important of the three, and, consequently, has been the most extensively developed. It is exposed at the surface in several good outcrops in the immediate vicinity of the portal and at several points to the north of the tunnel on the same side of the river. Several small stringers were found in the stream bed in a search for the extension of the ore-body across the river, but on the opposite bank no definite indication of its continuation had been found up to the time of this examination, though the search was being carried on with vigour. A tunnel, 720 feet in length at the time of the examination, had been driven along the strike of the ore-body and exposed, except for a short distance in from the portal, an almost continuous body of ore carrying between 3 per cent and 4 per cent copper and small values in gold. The

width of the zone has been tested by a crosscut at a point 240 feet in from the portal and also by two drill holes. The crosscut exposed 15 or 16 feet of ore averaging 3.7 per cent copper, and also a small zone 3 feet wide of 3.8 per cent ore separated from the larger body by 6 feet of barren rock. The diamond drill hole has proved this ore to a depth of 120 feet below the tunnel, and since the face of the tunnel is now about 440 feet vertically below the surface, the perpendicular dimension of the zone must be quite large. Though the extension of this zone across the river has not been proved, its continuation in the opposite direction for a considerable distance beyond the present face of the tunnel (which is in good ore) is highly probable. This zone is exposed also in a small crosscut tunnel 100 feet northeast of the portal of the River tunnel. This tunnel is on a continuation of the main ore-shoot of this zone and samples taken from it gave 2.5 per cent copper.

The Cave ore zone is more irregular, and, therefore, more difficult to follow and to outline. Its irregularity is probably due to the fact that it follows, in part at least, one of the irregular bedding planes between two flows of basalt, rather than the shear zones which are more uniform in strike and dip.

The tunnel was begun on an excellent surface showing and was driven 180 feet in a south 75 degrees east direction, the first 60 feet passing through ore averaging about 4 per cent copper. At 180 feet from the portal a crosscut was driven south 10 degrees west for a distance of 190 feet. The first 60 feet of the crosscut, 10 feet of which averaged 1.3 per cent copper, passed through low grade ore averaging 0.9 per cent copper. The next 20 feet encountered unmineralized rock and beyond this occurs 36 feet of ore averaging 2.58 per cent, one section of which averaged 5.1 per cent copper for a distance of 11 feet. For the remaining distance the tunnel passed through low grade ore.

At the point where the crosscut encountered the rich streak, a drift was driven south 60 degrees east for a distance of 200 feet, passing through good ore for the most part. It ends, however, in low grade ore, and small crosscuts driven short distances in both directions failed to locate any good ore. No work is being done at present on this ore-body.

About midway between the River and Cave zones outcrops of good ore, known as the Archibald showings, were discovered at an elevation of 260 feet above the levels of the tunnels. A diamond drill hole connecting the two tunnels underneath these showings proved the presence of 2 feet of 4 per cent ore and 10 to 12 feet of low grade material.

A crosscut was driven from the river tunnel to this ore and encountered better ore than was inferred from the values given by the drill cores. A drift a few feet in length has been driven on this ore.

Summary and Conclusions.

Geologic History and Origin of the Deposits. The salient facts brought out in the preceding description, and upon which the conclusions are based, are the following:

The deposits occur in the Metchosin formation, a series of basic lava flows and tuffs which were extruded in early Tertiary times.

These volcanic rocks were compressed into large folds striking north 60 degrees to 70 degrees west and have dips varying from zero to 85 degrees. The greater part of the formation exposed in the district forms the southern limb of a geosyncline and consequently dips to the northeast. The folding was accompanied by considerable fracturing in the less competent beds.

In the middle Tertiary the Metchosin formation was intruded by a number of stocks and dykes of coarse, basic, gabbro which appear to have invaded the volcanics along two principal axes of folding, the southern of which passes through the Sunloch property.

These intrusions had a profound metamorphic action on the volcanic rocks adjacent to them. A pronounced shearing was developed roughly parallel to the con-

tacts. The augite was replaced by hornblende so that great quantities of this mineral developed in the volcanics, and in parts of the gabbro itself, by solutions emanating from the interior parts of the gabbro and invading the volcanic rocks chiefly along shear and bedding planes.

Following these solutions were others which deposited more hornblende, actinolite, apfite, epidote, quartz, pyrite, molybdenite, and chalcopyrite, the last in sufficient quantity to form commercial ore-shoots.

In very recent times oxidation to a slight extent has taken place, producing small amounts of limonite and chalcocite.

Status of the Sunloch Property. It will be seen from the descriptions of these ore deposits that a considerable tonnage of commercial ore has already been proved on this property and that, as yet, only part of the possible ore-bearing ground has been tested.

In his report to the company at the end of 1918, the managing director, Mr. R. H. Stewart, estimated that from 100,000 to 150,000 tons of ore, averaging 3 per cent to 5 per cent, had up to that time been proved. This estimate was based on actual measurements and numerous assays and should be nearly correct. Since that time development work has been continued with some success so that the above quantity has been materially increased.

Regarding future possibilities, it is the writer's opinion that the ore-body exposed in the River tunnel will continue in a southeasterly direction for a considerable distance, possibly as far again, beyond the present face of the tunnel. The ore will in all probability continue to a great depth below the present workings, possibly until the gabbro is reached and beyond, though not without variations in size and tenor. It does not seem to continue in a northwesterly direction.

The ore-body in the Cave tunnel appears to cease at the face of the present workings, but since it is a very irregular ore-body it may be found to regain its strength at some distance beyond the present face. Good showings, which might prove to be a continuation of this ore-body, occur on the opposite side of the river. The Cave ore zone will persist with depth, but owing to its irregular nature may be difficult to follow.

The Archibald zone has possibilities, but not as great as the other two.

It must be borne in mind that this deposit belongs to a class which is commonly found in the coast section of British Columbia, i.e. the class which consists of disseminations or small veins of ore formed in volcanic rocks at the contacts of plutonic intrusives, and which have been frequently very disappointing. They usually consist of disseminated bodies that are too low grade, or of veins which are too small, the reasons being that the impervious nature of ordinary volcanic rocks, and their high degree of chemical inertness resist the tendency to replacement by the mineralizing solutions. These two disadvantages, however, have been to some extent removed in the case of the Sunloch deposit by the shearing which has made the volcanics more pervious and by the presence of such unusually large quantities of hornblende which lends itself to the process of replacement more readily than the minerals usually found in volcanic rocks. It is for these reasons that the ore deposits of this district are found to be invariably confined to sheared and hornblenditized rocks.

It must also be remembered that the formations associated with these deposits are not developed on a very large scale, being confined to only a few hundred square miles, and that, therefore, the ore deposits in them cannot be expected to rank with the great ore deposits associated with continent wide formations; and again that the other similar copper deposits associated with these formations have all, up to the present, been proved to be quite limited in extent. From these considerations and even

though there are 150,000 tons of ore proved already on the Sunloch property and another 100,000 tons of probable ore, yet one would expect the deposit to have a tonnage measured in the hundreds of thousands or a few millions at most. In other words it is a good deposit of moderate rather than great size.

COQUIHALLA MAP-AREA, B.C.

By Charles Camsell.

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INTRODUCTION.

The Coquihalla map-area comprises a strip of country extending from the town of Hope on Fraser river up the valley of Coquihalla river as far as Boston Bar creek and takes in a belt from 4 to 10 miles deep on either side of the river.

The Kettle Valley railway runs through this strip of country and all the territory embraced by the map-area is tributary to that railway.

A topographic base map of this area was made by F. S. Falconer in the season of 1918, on a scale of 1 mile to 1 inch with contour intervals of 100 feet.

Owing to the pressure of other duties it was found impossible to spend more than four weeks in the field, on geological mapping, and, since the area is very difficult to cover on account of high relief, dense forest, and lack of trails, only a very small part of the work necessary to a full report was done.

Placer gold has been mined in early days on Coquihalla river and more recently lode gold deposits have been developed that are very high grade.

Other deposits of gold, silver, lead, and copper occur in territory immediately adjacent to the area, namely, on Silver creek, Siwash creek, Tulameen river, and elsewhere; but the area itself is distinctly a gold-bearing one and deposits of this metal are likely to be the most important developed in it.

The mapping of the area was undertaken with a view to providing accurate structural and other geological data on which plans for prospecting and development might be based. At the same time prospectors have frequently been urged by the writer to confine these efforts to districts in which transportation either by rail or water is already provided and to do intensive detailed prospecting in such districts rather than to go into more remote regions where little or no transportation facilities exist.

Coquihalla area is a district where rail transportation has only recently been provided, by the Kettle Valley railway, but which has remained unprospected up to recent years owing to the physical difficulties that confront the prospector in any attempts to carry on his work.

The limited knowledge of the geology of the area was, however, sufficient to recommend it to the prospector because of the occurrence of gold-bearing quartz veins in the adjacent Siwash Creek district where the same association of geological formations occurs as in the Coquihalla area.